

CHAT SYSTEM

1.REGISTER PAGE:

- Name input
- Email input
- Password input
- Register button
- Click to login

2.LOGIN PAGE:

- Name input
- Email input
- Password input
- Click to login

3.AUTHENTICATION FLOW :

REGISTRATION : react , post/register, bcrypt hashpassword stores ,mongodb , success.

LOGIN : react , post/login, express, mongodb ,checks bcrypt hash, React.

4.CHAT PAGE:

LEFT SIDEBAR / HAMBURGER SYMBOL:

- Add a team
- Manage accounts
- Current user
- Logout

LANDING PAGE UI:

- Chats and conversations
- Get api to load the chats and conversations
- User open a colleague chat,
- Get api to bring the previous conversations history.
- Backend returns the message.
- Frontend renders it.

5. HOW SENDING A MESSAGE WORKS:

- User types an hello message and click send.
- React send an socket.io
- socket.io verifies the authenticated user and the receiver.
- Server creates a message document and save it to mongodb
- Receiver react app receives the message and updates in the UI.

6.LOGOUT

User clicks logout
Jwt connections is disabled.
Socket connection is also disabled.
Server disconnects and mark the user offline.
React rendirects it as an login.

7.MONGODB DATA COLLECTION:

```
{  
  Id: 123456  
  Name:ashwin  
  Email:ashwin@gmail.com  
  Password:  
  Status:offline  
  Lastseen: 10:30  
  createdAt:  
}
```

8.GROUP CHATS PAGE:

Create group
Select members,
Api groups
Express
validate members
Create group document
Mongodb saves.

HOW MONGODB STORES IT:

```
{  
  _id: ObjectId("group123"),  
  name: "Project Team",  
  createdBy: ObjectId("abinayald")  
  members: [  
    ObjectId("abinayald"),  
    ObjectId("ravild"),  
    ObjectId("vijayld"),  
    ObjectId("priyald")  
  ],  
  createdAt: Date  
}
```

9.CHANNELS PAGE:

Admin clicks create a channel
Channel name
Description
Select private/public channel
Post api/channel
Express
Creates channel
Mongodb stores
Return channel
React displays it

CHANNELS MONGODB STRUCTURE:

```
{
  _id: ObjectId("channel123"),
  name: "frontend",
  description: "Frontend development",
  workspaceId: ObjectId("workspace123"),
  type: "public",
  createdBy: ObjectId("adminId"),
  members: [
    ObjectId("abinayId"),
    ObjectId("ravild"),
    ObjectId("vijayId")
  ],
  createdAt: Date
}
```

USER OPENS A CHANNEL PAGE;

Click backend
React gets channel id
Get api/channels/channelid
Express
Mongodb
Find messages with channel id
Returns messages
React displays